

WEB TEXNOLOGIYALAR

LABORATORIYA ISHI ■ 14

Mavzu: Progressive Web App (PWA) asoslari: oddiy PWA yaratish

1. Ishning maqsadi

- Progressive Web App (PWA) tushunchasi va afzalliklarini o'rganish
- Web App Manifest fayli yaratish va sozlash
- Service Worker — oflayn ishlash mexanizmini o'zlashtirish
- Cache API yordamida resurslarni keshlashtirish
- PWA ni mobil qurilmaga o'rnatish (Add to Home Screen)
- Push Notification asoslarini o'rganish
- Lighthouse vositasi bilan PWA sifatini tekshirish

2. Nazariy qism

PWA nima?

Progressive Web App (PWA) — veb-sayt va mobil ilova xususiyatlarini birlashtirgan zamonaviy veb-texnologiya. PWA odatdagi brauzerda ochiladi, lekin mobil ilovadek qurilmaga o'rnatiladi, oflayn ishlaydi va push bildirishnomalar yuboradi. Google tomonidan 2015-yilda taqdim etilgan bu yondashuv bugungi kunda Twitter, Pinterest, Uber, Starbucks kabi yirik kompaniyalar tomonidan qo'llaniladi.

PWA asosiy xususiyatlari:

Xususiyat	Tavsifi
Progressiv	Har qanday brauzerda ishlaydi — eski va yangisida ham
Responsive	Telefon, planshet, kompyuter — barcha ekranlarga moslashadi
Oflayn ishlash	Service Worker orqali internet bo'lmasa ham ishlaydi
O'rnatiladi	Qurilmaga ilova sifatida o'rnatiladi (App Store siz)
Tez ishlaydi	Kesh tufayli tez yuklanadi, siliq animatsiyalar
Xavfsiz	Faqat HTTPS orqali ishlaydi
Push xabarnoma	Foydalanuvchiga bildirishnoma yuborish mumkin
Yangilanadi	Har doim so'nggi versiyani ko'rsatadi

PWA vs Oddiy sayt vs Native ilova:

Xususiyat	Oddiy sayt	PWA	Native ilova
O'rnatish	Shart emas	Ixtiyoriy	Majburiy
App Store	Yo'q	Yo'q	Kerak
Oflayn ishlash	Yo'q	Ha	Ha
Push xabarnoma	Yo'q	Ha	Ha

Yangilanish	Avtomatik	Avtomatik	Qo'lda
Qurilma imkonlari	Cheklangan	O'rta	To'liq
Ishlab chiqish	Oson	O'rta	Qiyin

PWA texnik asoslari:

- Web App Manifest (manifest.json) — ilova nomi, ikonka, rang va boshqa meta-ma'lumotlar
- Service Worker (sw.js) — brauzer va server orasida ishlaydigan proksi skript
- Cache API — resurslarni (HTML, CSS, JS, rasm) kesh xotirasida saqlash
- HTTPS — xavfsiz ulanish (Service Worker faqat HTTPS da ishlaydi)
- IndexedDB — oflayn rejimda ma'lumotlarni saqlash uchun baza

3. Loyiha tuzilishi

```
pwa-ilova/
index.html <- Asosiy sahifa
style.css <- Uslublar
app.js <- Asosiy JavaScript (SW ro'yxatdan o'tkazish)
sw.js <- Service Worker (kesh va oflayn mantiq)
manifest.json <- Web App Manifest (ilova meta-ma'lumotlari)
icons/
icon-192.png <- Ilova ikonkasi (192x192 px)
icon-512.png <- Ilova ikonkasi (512x512 px)
```

4. Amaliy qism — bosqichma-bosqich bajarish

1-bosqich: manifest.json — ilova manifesti

Manifest fayli brauzerga ilova haqida ma'lumot beradi: nomi, ikonkasi, rangi va ekranda qanday ko'rinishi.

```
{
  "name": "Mening PWA Ilovam",
  "short_name": "PWA Ilova",
  "description": "Oddiy PWA – oflayn ishlaydigan veb-ilova",
  "start_url": "/index.html",
  "display": "standalone",
  "orientation": "portrait",
  "background_color": "#ffffff",
  "theme_color": "#2196F3",
  "lang": "uz",
  "icons": [
    {
      "src": "icons/icon-192.png",
      "sizes": "192x192",
      "type": "image/png",
```

```

"purpose": "any maskable"
},
{
  "src": "icons/icon-512.png",
  "sizes": "512x512",
  "type": "image/png",
  "purpose": "any maskable"
}
]
}

```

display qiymati	Tavsifi
standalone	Ilova mustaqil oyna sifatida ochiladi (tavsiya etiladi)
fullscreen	To'liq ekranda ochiladi (o'yinlar uchun)
minimal-ui	Minimal brauzer boshqaruvi bilan ochiladi
browser	Odatdagi brauzer rejimida ochiladi

2-bosqich: index.html — asosiy sahifa

```

<!DOCTYPE html>
<html lang="uz">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="theme-color" content="#2196F3">
  <meta name="description" content="Oddiy PWA ilova">

  <!-- Manifest ulash -->
  <link rel="manifest" href="manifest.json">

  <!-- iOS uchun -->
  <meta name="apple-mobile-web-app-capable" content="yes">
  <meta name="apple-mobile-web-app-title" content="PWA Ilova">
  <link rel="apple-touch-icon" href="icons/icon-192.png">

  <link rel="stylesheet" href="style.css">
  <title>Mening PWA Ilovam</title>
</head>
<body>
  <header>
    <h1>Mening PWA Ilovam</h1>
    <p id="status">Onlayn</p>
  </header>

  <main>

```

```

<div class="card">
<h2>Xush kelibsiz!</h2>
<p>Bu ilova oflayn rejimda ham ishlaydi.</p>
</div>

<div class="card">
<h2>Eslatmalar</h2>
<input type="text" id="eslatma" placeholder="Eslatma kiriting...">
<button onclick="eslatmaQosh()">Qo'shish</button>
<ul id="royxat"></ul>
</div>

<button id="oRnatBtn" style="display:none">
Ilovani o'rnatish
</button>
</main>

<script src="app.js"></script>
</body>
</html>

```

3-bosqich: style.css — uslublar

```

* { box-sizing: border-box; margin: 0; padding: 0; }

body {
font-family: 'Segoe UI', sans-serif;
background: #f5f5f5;
color: #333;
}

header {
background: #2196F3;
color: white;
padding: 16px 20px;
display: flex;
justify-content: space-between;
align-items: center;
position: sticky;
top: 0;
}

main { padding: 20px; max-width: 600px; margin: 0 auto; }

.card {
background: white;
border-radius: 8px;
padding: 20px;
}

```

```
margin-bottom: 16px;
box-shadow: 0 2px 8px rgba(0,0,0,0.1);
}

input {
width: 100%;
padding: 10px;
border: 1px solid #ddd;
border-radius: 6px;
margin: 8px 0;
font-size: 16px;
}

button {
background: #2196F3;
color: white;
border: none;
padding: 10px 20px;
border-radius: 6px;
cursor: pointer;
font-size: 16px;
width: 100%;
margin-top: 8px;
}

button:hover { background: #1976D2; }

#status {
padding: 4px 10px;
border-radius: 12px;
font-size: 13px;
background: #4CAF50;
}

#status.offlayn { background: #f44336; }
```

4-bosqich: app.js — Service Worker ro'yxatdan o'tkazish

```
// Service Worker ro'yxatdan o'tkazish
if ('serviceWorker' in navigator) {
window.addEventListener('load', () => {
navigator.serviceWorker.register('/sw.js')
.then(reg => {
console.log('Service Worker ro'yxatdan o'tdi:', reg.scope);
})
.catch(err => {
console.log('Service Worker xatosi:', err);
})
})
}
```

```

});
});
}

// Onlayn / Oflayn holat
const statusEl = document.getElementById('status');

function holatniYangilash() {
  if (navigator.onLine) {
    statusEl.textContent = 'Onlayn';
    statusEl.className = '';
  } else {
    statusEl.textContent = 'Oflayn';
    statusEl.className = 'oflayn';
  }
}

window.addEventListener('online', holatniYangilash);
window.addEventListener('offline', holatniYangilash);
holatniYangilash();

// Eslatmalar (localStorage)
function eslatmaQosh() {
  const input = document.getElementById('eslatma');
  const matn = input.value.trim();
  if (!matn) return;

  const eslatmalar = JSON.parse(localStorage.getItem('eslatmalar') || '[]');
  eslatmalar.push(matn);
  localStorage.setItem('eslatmalar', JSON.stringify(eslatmalar));
  input.value = '';
  eslatmalarniKorsat();
}

function eslatmalarniKorsat() {
  const eslatmalar = JSON.parse(localStorage.getItem('eslatmalar') || '[]');
  const ul = document.getElementById('royxat');
  ul.innerHTML = eslatmalar.map(e =>
    `<li style="padding:6px 0;border-bottom:1px solid #eee">${e}</li>`
  ).join('');
}

eslatmalarniKorsat();

// O'rnatish tugmasi (Add to Home Screen)
let deferredPrompt;
const oRnatBtn = document.getElementById('oRnatBtn');

window.addEventListener('beforeinstallprompt', (e) => {

```

```
e.preventDefault();
deferredPrompt = e;
oRnatBtn.style.display = 'block';
});

oRnatBtn.addEventListener('click', async () => {
  if (!deferredPrompt) return;
  deferredPrompt.prompt();
  const result = await deferredPrompt.userChoice;
  console.log('Foydalanuvchi tanlovi:', result.outcome);
  deferredPrompt = null;
  oRnatBtn.style.display = 'none';
});
```

5-bosqich: sw.js — Service Worker (kesh va oflayn)

Service Worker — brauzer orqasida ishlaydigan alohida JavaScript fayli. U tarmoq so'rovlarini ushlab, keshdan javob qaytara oladi.

```
const KESH_NOMI = 'pwa-kesh-v1';
const KESH_FAYLLAR = [
  '/',
  '/index.html',
  '/style.css',
  '/app.js',
  '/manifest.json',
  '/icons/icon-192.png',
  '/icons/icon-512.png'
];

// INSTALL – barcha fayllarni keshga olish
self.addEventListener('install', (event) => {
  console.log('Service Worker o'rnatilmoqda...');
  event.waitUntil(
    caches.open(KESH_NOMI)
      .then(kesh => {
        console.log('Fayllar keshga olinmoqda');
        return kesh.addAll(KESH_FAYLLAR);
      })
      .then(() => self.skipWaiting())
  );
});

// ACTIVATE – eski keshlarni tozalash
self.addEventListener('activate', (event) => {
  console.log('Service Worker faollashtirildi');
```

```

event.waitUntil(
  caches.keys().then(keshNomlar => {
    return Promise.all(
      keshNomlar
        .filter(nom => nom !== KESH_NOMI)
        .map(nom => caches.delete(nom))
    );
  }).then(() => self.clients.claim())
);

// FETCH – so'rovlarni ushlab
self.addEventListener('fetch', (event) => {
  event.respondWith(
    caches.match(event.request)
      .then(keshJavob => {
        // Keshda bor – keshdan qaytarish
        if (keshJavob) {
          return keshJavob;
        }
        // Keshda yo'q – tarmoqdan olish
        return fetch(event.request)
          .then(tarmoqJavob => {
            // Yangi resursni keshga saqlash
            const keshNusxa = tarmoqJavob.clone();
            caches.open(KESH_NOMI)
              .then(kesh => kesh.put(event.request, keshNusxa));
            return tarmoqJavob;
          })
          .catch(() => {
            // Tarmoq yo'q va keshda ham yo'q
            return new Response(
              '<h2>Internet aloqasi yo'q</h2>',
              { headers: { 'Content-Type': 'text/html; charset=utf-8' } }
            );
          });
      })
  );
});

```

5. Service Worker hodisalari

Hodisa	Qachon ishlaydi	Vazifasi
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install	SW birinchi marta o'rnatilganda	Kerakli fayllarni keshga olish
activate	SW faollashganda	Eski keshlarni o'chirish
fetch	Har bir tarmoq so'rovida	Kesh yoki tarmoqdan javob qaytarish
push	Server xabarnoma yuborganda	Push bildirishnoma ko'rsatish
sync	Internet qayta ulanganda	Oflayn paytdagi amallarni yuborish

6. Lighthouse bilan PWA tekshirish

Lighthouse — Google Chrome DevTools ichidagi avtomatik sifat tekshiruv vositasi. PWA, Performance, Accessibility va SEO bo'yicha ball beradi.

- Chrome DevTools ni oching (F12)
- Lighthouse bo'limiga o'ting
- Progressive Web App ni belgilang
- Analyze page load tugmasini bosing
- Natijada 100 ball olishga harakat qiling

Lighthouse mezonlari	Tavsifi
Installable	manifest.json va Service Worker mavjud
PWA Optimized	HTTPS, offline sahifa, meta viewport bor
Performance	Sahifa tez yuklanadi (LCP, FID, CLS)
Accessibility	Barcha foydalanuvchilar uchun qulay

7. Topshiriq

Mavzu: Ob-havo PWA — oflayn ishlaydigan oddiy ob-havo ilovasi yarating.

Bajarish bosqichlari:

1-bosqich: Loyiha papkasini yarating: manifest.json, sw.js, index.html, style.css, app.js

2-bosqich: manifest.json da ilova nomini, ikonkalarni va theme_color ni sozlang

3-bosqich: index.html da asosiy sahifa tuzilmasi (shahar nomi, harorat ko'rsatgich) yarating

4-bosqich: app.js da Service Worker ni ro'yxatdan o'tkazing va online/offline holatni boshqaring

5-bosqich: sw.js da install, activate va fetch hodisalarini yozing

6-bosqich: localStorage da oxirgi ob-havo ma'lumotini saqlang — oflayn ko'rsatish uchun

7-bosqich: VS Code Live Server orqali ishga tushiring va Chrome DevTools bilan tekshiring

8-bosqich: Lighthouse da PWA ballingizni o'lchang va skrinshotini oling

8. Natija va Xulosa

Laboratoriya davomida quyidagilar bajarildi:

- PWA tushunchasi, xususiyatlari va texnik asoslari o'rganildi
- manifest.json fayli yaratildi va ilova meta-ma'lumotlari sozlandi

- index.html sahifasiga manifest va meta teglar ulandi
- Service Worker (sw.js) yozildi — install, activate, fetch hodisalari
- Cache API orqali barcha resurslar keshlantirildi
- Onlayn/oflayn holat real vaqtda ko'rsatildi
- localStorage yordamida eslatmalar oflayn saqlandi
- Add to Home Screen (o'rnatish) tugmasi qo'shildi
- Lighthouse vositasi bilan PWA sifati baholandi

Xulosa: Progressive Web App texnologiyasi veb-saytlarga mobil ilova imkoniyatlarini beradi. Service Worker oflayn ishlashni ta'minlaydi, manifest.json esa ilovani qurilmaga o'rnatish imkonini beradi. PWA App Store ga joylashtirishsiz foydalanuvchiga yetib boradi va barcha platformalarda (Android, iOS, Windows) ishlaydi. Bu texnologiya kelajakda veb va native ilovalar o'rtasidagi chegarani yanada yo'qotib boradi.